2. A die is thrown twice and the sum of the number appearing is observed to be 6. What is the conditional probability that the number 4 has appeared at least once?

3. An unbiased die is thrown twice. Let the event A be odd number on the first throw and B the event odd number on the second throw. Check whether A and B events are independent.

4. Probability of solving specific problem independently by A and B are $\frac{1}{2}$ and $\frac{1}{3}$ respectively. If both try to solve the problem independently, find the probability that the problem is

   (i) solved

   (ii) exactly one of them solves the problem

5. Suppose one person is selected at random from a group of 100 persons are given in the following

Title	Psychologist	Socialist	Democrate	Total
Men	15	25	10	50
Women	20	15	15	50
Total	35	40	25	100

   What is the probability that the man selected is a Psychologist?

6. Two urns contains the set of balls as given in the following table

Title	White	Red	Black
Urn 1	10	6	9
Urn 2	3	7	15

   One ball is drawn from each urn and find the probability that

   (i) both balls are red

   (ii) both balls are of the same colour.

7. Bag I contains 3 Red and 4 Black balls while another Bag II contains 5 Red and 6 Black balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that it was drawn from Bag I.

8. The first of three urns contains 7 White and 10 Black balls, the second contains 5 White and 12 Black balls and third contains 17 White balls and no Black ball. A person chooses an urn at random and draws a ball from it. And the ball is found to be White. Find the probabilities that the ball comes from

   (i) the first urn

   (ii) the second urn

   (iii) the third urn

9. Three boxes B_1, B_2, B_3 contain lamp bulbs some of which are defective. The defective proportions in box B_1, box B_2 and box B_3 are respectively $\frac{1}{2}$, $\frac{1}{8}$ and $\frac{3}{4}$. A box is selected at random and a bulb drawn from it. If the selected bulb is found to be defective, what is the probability that box B_1 was selected?

10. Three horses A, B, C are in race. A is twice as likely to win as B and B is twice as likely to win as C. What are their respective probabilities of winning?

11. A die is thrown. Find the probability of getting

    (i) a prime number

    (ii) a number greater than or equal to 3

12. Ten cards numbered 1 to 10 are placed in a box, mixed up thoroughly and then one card is drawn randomly. If it is known that the number on the drawn card is

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